DDRAM-O Milestone 1 Specification

Simplified Orthogonal-View Device and Timing Model

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Status. This document defines the first implementable DDRAM-O model for Ramulator2. Architectural interfaces and state transitions are normative. Electrical timing values introduced specifically for DDRAM-O are placeholders and must remain configurable until replaced by SPICE-calibrated values.

Contents

1 Purpose and Scope 1
- 1.1 Normative language . . . 1

2 Goals and Non-Goals 1
- 2.1 Milestone 1 goals . . . 1
- 2.2 Explicit non-goals . . . 1

3 Architectural Model 1
- 3.1 Views and addressing . . . 1
- 3.2 Per-bank state . . . 2
- 3.3 Simplified concurrency rule . . . 2

4 Command Interface 2
- 4.1 Required commands . . . 2
- 4.2 Multi-row activation abstraction . . . 2
- 4.3 Reserved primitive timings . . . 2

5 Timing Model 2
- 5.1 Parameters and initial values . . . 2
- 5.2 Provisional timing ledger and rationale . . . 2
- 5.3 Activation-like command class . . . 2
- 5.4 Required timing relationships . . . 2
- 5.5 Activation power window . . . 2

6 Prerequisites and State Transitions 2
- 6.1 Conventional requests . . . 2
- 6.2 Atomic multi-row activation . . . 3

7 Configuration Interface 3

8 Trace and Statistics Requirements 3

9 Verification Plan 3

10 Calibration and Evolution 3
- 10.1 SPICE calibration . . . 3
- 10.2 Intersection-aware extension . . . 4
- 10.3 Functional PuD extension . . . 4

11 Implementation Boundary 4

1 Purpose and Scope

DDRAM-O provides two access structures over the same DRAM cell array. View A uses the conventional row orientation; view B is rotated by 90°, so a row in one view geometrically corresponds to a column in the other. The complete architecture creates cell-intersection hazards between simultaneously open rows. Milestone 1 intentionally omits those hazards and models the two row paths as independent, subject to an explicit cross-view activation delay.

The purpose of this milestone is to establish a stable device interface for:

- view-specific conventional accesses;
- overlap between activity on the two row paths;
- atomic processing-using-DRAM (PuD) primitives;
- configurable, visible placeholder timings; and
- command traces suitable for correctness and scheduling analysis.

This is a timing and state model. Functional arithmetic, operand-row allocation, energy calibration, and operation scheduling are later milestones.

1.1 Normative language

The words **must**, **must not**, **should**, and **may** describe implementation requirements. A requirement marked **Provisional** defines the initial model but may be replaced by calibrated data without changing the command interface.

2 Goals and Non-Goals

2.1 Milestone 1 goals

1. Introduce a distinct `DDRAM_O` DRAM standard.
2. Preserve the conventional DDR5 access path through view A.
3. Permit one conventional row to remain open independently in each view.
4. Introduce explicit activation, access, and precharge commands per view.
5. Introduce an atomic multi-row activation primitive per view.
6. Permit cross-view overlap after a configurable minimum delay.
7. Preserve conventional command-, column-, and data-bus sharing.
8. Make every DDRAM-O-specific timing overridable from Python configuration.

2.2 Explicit non-goals

Milestone 1 does *not* model:

- A/B row-intersection conflicts or simultaneous-sensing hazards;
- masked intersection bits or half-amplitude sensing;
- persistent states with three or five conventionally open rows;
- functional data values or arithmetic correctness;
- compute-row allocation, operand lifetime, or spilling;
- calibrated multi-row activation current or energy; or
- end-to-end CPU trace offloading.

Results produced by this model must not be described as SPICE-calibrated.

3 Architectural Model

3.1 Views and addressing

View A is the host-oriented conventional view. View B is the orthogonal view. Both commands use the existing hierarchical Ramulator address vector. In Milestone 1, the row field is interpreted in the coordinate system of the selected view. Consequently, row number r in A and row number r in B do not denote the same geometric row.

More precisely, the row field is a *view-local line index*. For example, `A:10` names line 10 in the A coordinate system, whereas `B:10` names line 10 in the orthogonal B coordinate system. The equal integers are labels in different namespaces; they do not mean that one physical row has been opened twice. The two

selected lines may physically intersect at a cell, but Milestone 1 assigns that intersection no additional state transition, timing penalty, masking rule, or data-value effect. Thus both activations are legal in the simplified timing model. A later intersection-aware functional model must determine the shared-cell mapping and any resulting electrical or correctness restrictions.

The address mapper is not required to perform a physical transpose in this milestone. The selected command identifies the coordinate system. A later functional model will map both coordinate systems onto shared cells.

3.2 Per-bank state

Each physical bank maintains the state shown in Table 1.

Table 1: Required simplified state per physical bank.

Field	Domain	Meaning
`active_row[A]`	closed or row ID	Conventional row open through view A.
`active_row[B]`	closed or row ID	Conventional row open through view B.
`busy_until[A]`	cycle	Earliest cycle at which another A row-path operation may start.
`busy_until[B]`	cycle	Earliest cycle at which another B row-path operation may start.

The two active-row fields are independent. Opening or precharging one view must not change the other view's state. No equality check between the two row IDs is performed. A bank is conventionally closed only when both fields are closed.

3.3 Simplified concurrency rule

Commands on different views may overlap once their activation-like command starts are separated by `nXACTO`. Commands still contend for all shared resources inherited from DDR5. In particular, DDRAM-O adds row-path parallelism, not an additional off-chip data interface.

4 Command Interface

4.1 Required commands

Table 2 defines the Milestone 1 command set. "Local" means the state or timing effect applies only to the command's selected view.

4.2 Multi-row activation abstraction

`MRA_A` and `MRA_B` represent the complete sequence

$$\text{multi-row activation} \rightarrow \text{sense/majority} \rightarrow \text{restore} \rightarrow \text{precharge}. \tag{1}$$

The command must begin with its local view closed. If that view is open, its prerequisite is the corresponding local precharge. The opposite view may remain open. At command issue, the local view enters a timed busy state but does not acquire a persistent active-row identity. It is logically closed after the atomic operation.

Milestone 1 may use the command address as a representative bank/subarray target. The operand-row list belongs to the compound PuD request introduced in Milestone 2 and must not be encoded through unrelated address-vector fields.

4.3 Reserved primitive timings

The implementation should also define `nCOPY` and `nNOT` timing parameters now, even if explicit `COPY_A/B` and `NOT_A/B` commands are introduced with functional primitives in a later milestone. Reserving separate parameters prevents future calibration from coupling logically different primitives merely because their initial values are equal.

5 Timing Model

5.1 Parameters and initial values

All values are integer DRAM cycles. Table 3 distinguishes inherited DDR5 timings from DDRAM-O-specific placeholders. The current provisional timing ledger is revision M1-P0. This identifier and the resolved values should accompany experiment results until calibrated parameters replace the ledger.

The default `nXACTO` equals the current provisional DDRAM-P cross-view delay for comparability; it is not an electrical claim. The `nMRA` and `nCOPY` formulas follow the initial AP/AAP cost interpretation already used by the local PuD modeling infrastructure. SPICE calibration may replace the values, but should not rename the parameters or commands.

5.2 Provisional timing ledger and rationale

Table 4 records why each placeholder exists and what evidence is required to retire it. The defaults are modeling hypotheses, not predictions of a fabricated DDRAM-O device.

The implementation resolves the DDR5 preset first and then derives these values. User overrides take precedence and must be positive integer cycles. For the current `DDR5_3200AN` preset, `nRAS`= 52 and `nRP`= 24, yielding `nMRA`= 76 and `nCOPY`=`nNOT`= 82 cycles; `nXACTO` remains 8 cycles. The reserved `nCOPY` and `nNOT` values do not yet gate an issued command and therefore must not be interpreted as measured operation latency in Milestone 1 results.

5.3 Activation-like command class

For cross-view constraints, the activation-like class is

$$\mathcal{A}_A = \{\texttt{ACT_A}, \texttt{MRA_A}\},$$
$$\mathcal{A}_B = \{\texttt{ACT_B}, \texttt{MRA_B}\}.$$

Future copy and inversion commands must join the class if their physical implementation begins with activation of their selected view.

5.4 Required timing relationships

An `MRA` command does not occupy the data bus. Opposite-view row-path commands may overlap its local occupancy after satisfying `nXACTO`, but ordinary column commands remain subject to shared DDR5 constraints.

5.5 Activation power window

Provisional Each `MRA` counts as one activation event for `nFAW` and the appropriate `nRRD` family. This is a bookkeeping approximation, not a claim that a multi-row activation draws the current of one ordinary activation. A later calibrated model may introduce a weighted activation budget or a dedicated constraint.

6 Prerequisites and State Transitions

6.1 Conventional requests

For a request assigned to view V:

1. If the target row is open in V, issue the terminal column command.
2. If V is closed, issue `ACT_`V, then the column command.
3. If another row is open in V, issue `PRE_`V, `ACT_`V, and then the column command.
4. State in the other view is neither a prerequisite nor a victim.

Table 2: Required DDRAM-O commands.

Command	Scope	Semantics
`ACT_A`	Bank, local	Open the target row in view A.
`ACT_B`	Bank, local	Open the target row in view B.
`PRE_A`	Bank, local	Close only the row open in view A.
`PRE_B`	Bank, local	Close only the row open in view B.
`RD_A`, `WR_A`	Bank, shared data path	Read or write the row open in view A.
`RD_B`, `WR_B`	Bank, shared data path	Read or write the row open in view B.
`MRA_A`	Bank/subarray, local	Execute an atomic multi-row activation on view A; no persistent multi-row-open state remains.
`MRA_B`	Bank/subarray, local	Symmetric atomic multi-row activation on view B.
`PREab`	Rank	Close both views in all targeted banks, for maintenance compatibility.
`PREsb`	Same-bank scope	Close both views in the targeted banks, for maintenance compatibility.

Table 3: DDRAM-O-specific timing parameters.

Parameter	Initial value	Status	Interpretation
`nXACTO`	8	Provisional	Minimum start-to-start delay between activation-like commands on opposite views of one bank/subarray.
`nMRA`	`nRAS` + `nRP`	Provisional	Local-view occupancy of an atomic multi-row activation, including restoration and precharge.
`nCOPY`	`nRAS` + `nRP` + $\lceil$`nRP`$/4\rceil$	Provisional	Initial SIMDRAM-style activate–activate–precharge copy cost.
`nNOT`	`nCOPY`	Provisional	Initial inversion cost when represented as a DCC-assisted copy.

6.2 Atomic multi-row activation

For `MRA_`V:

1. If V is open, issue `PRE_`V first.
2. The other view need not be closed.
3. On issue, set `busy_until[`V`]` to the current cycle plus `nMRA`.
4. Leave `active_row[`V`]` closed.
5. Do not change the other view's active row or busy deadline.

The controller must preserve an operation's selected view after its first prerequisite issues, following the same commitment principle used for DDRAM-P.

7 Configuration Interface

The Python DRAM specification must accept explicit overrides. A representative configuration is:

```
ramulator.dram.DDRAM_O(
    org_preset="DDR5_16Gb_x8",
    timing_preset="DDR5_4800AN",
    rank=1,
    nXACTO=8,
    # Other DDRAM-O timings use derived defaults.
)
```

Derived defaults should be resolved after the inherited DDR5 preset so that changing `nRAS` or `nRP` changes the placeholder consistently. Every experiment result must record:

- the resolved values of `nXACTO`, `nMRA`, `nCOPY`, and `nNOT`;
- the DDR5 organization and timing presets;
- a timing-model revision string, initially `M1-P0`; and
- whether intersection modeling is enabled, initially `false`.

Invalid negative or zero DDRAM-O-specific timing values must be rejected after default resolution.

8 Trace and Statistics Requirements

The existing trace recorder must represent every DDRAM-O command without changing conventional trace semantics. Each event should expose:

- command name and issue cycle;
- selected view;
- bank/subarray and representative row address;
- whether it is a conventional or PuD command;
- the timing constraint that determined readiness; and
- the resolved timing value used by that constraint.

The initial statistics should include counts of view-A and view-B activations, multi-row activations by view, cycles each view is busy, cross-view overlap cycles, and stalls attributed to `nXACTO` or `nMRA`.

9 Verification Plan

Milestone 1 is complete only when the following tests pass.

Every timing test should independently replay the issued history through the device prerequisite and timing checks, as the DDRAM-P suite already does.

10 Calibration and Evolution

The command interface is intended to remain stable while the physical model becomes more detailed.

10.1 SPICE calibration

The first calibration step replaces `nXACTO` using the minimum safe separation between opposite-view sensing events under the selected voltage, temperature, process corner, and row multiplicity. The report should provide the time in nanoseconds and the conversion rule

$$\texttt{nXACTO} = \left\lceil \frac{t_{\mathrm{XACTO}}}{t_{\mathrm{CK}}} \right\rceil . \tag{2}$$

Multi-row activation simulations should subsequently calibrate local-view occupancy, restoration time, current limits, and whether MRA should carry more than one activation unit in the power window.

Each calibration update must increment the timing-ledger revision, retain the superseded defaults in version history, and record

Table 4: Provisional timing ledger, revision M1-P0.

Item	Current rule	Reason for the choice	Replacement evidence
nXACTO	8 cycles	Matches the DDRAM-P cross-view placeholder, giving initial A/B comparisons the same assumed coupling delay. It is deliberately independent of the DDR5 speed bin.	Worst-case SPICE separation between opposite-view sensing starts, converted with $\lceil t/t_{CK} \rceil$.
nMRA	nRAS + nRP	Treats atomic MRA as one active/restore interval followed by complete local precharge. It is conservative in state lifetime but adds no unsupported circuit-specific phase.	Simulated multi-row sensing, restoration, and precharge completion time for each supported row multiplicity.
nCOPY	nMRA + $\lceil$nRP/4$\rceil$	Starts from MRA occupancy and reserves a small explicit margin for the additional internal phase of an activate–activate–precharge copy. Keeping it distinct enables later calibration without changing the interface.	End-to-end source-to-destination copy waveform, including restoration and safe reuse of the local view.
nNOT	nCOPY	Models the initial DCC-assisted inversion path as copy-equivalent when no independent inversion measurement is available. Equality is not a permanent architectural requirement.	Dedicated inversion-path latency, or evidence that the copy and inversion paths are identical.
MRA CA occupancy	2 cycles	Conservatively matches DDR5 address-bearing activation commands while the final multi-row operand encoding is unspecified.	Final command encoding and command/address bus protocol.
MRA activation weight	1 event	Lets existing nRRD/nFAW machinery limit MRA immediately, without claiming that MRA draws ordinary-ACT current.	Calibrated current envelope and either a weighted activation budget or dedicated MRA power constraint.

Table 5: Normative timing relationships.

Preceding	Following	Delay	Scope
$\mathcal{A}_A$	$\mathcal{A}_B$	nXACTO	Same bank/subarray, cross-view
$\mathcal{A}_B$	$\mathcal{A}_A$	nXACTO	Same bank/subarray, cross-view
MRA_A	Any A row command	nMRA	Local view
MRA_B	Any B row command	nMRA	Local view
ACT_A/B	Local RD/WR	nRCD	Inherited DDR5 behavior
ACT_A/B	Local PRE	nRAS	Inherited DDR5 behavior
PRE_A/B	Local activation-like	nRP	Local view
RD_A/B, WR_A/B	Column commands	DDR5 column constraints	Shared bank/channel data path

the circuit configuration, process/voltage/temperature corner, row multiplicity, time in nanoseconds, clock period, rounding rule, and resulting cycle count. This makes results produced with provisional and calibrated models distinguishable.

10.2 Intersection-aware extension

A later revision may add intersection identity, masking, and simultaneous sensing restrictions. It should extend readiness checks and functional data mapping, not replace view-specific command names or provisional timing parameters. Experiments must report which revision they use.

10.3 Functional PuD extension

Milestone 2 will attach explicit operand and destination rows to compound PuD requests. A functional shadow array will map the orthogonal coordinate systems onto shared cells and validate majority, addition, and multiplication. The atomic MRA_A/B commands defined here remain the timing-level leaves of those operation graphs.

11 Implementation Boundary

The recommended separation of responsibilities is:

- **DRAM specification:** commands, state transitions, prerequisites, timing constraints, and parameter resolution;
- **controller:** request preparation, selected-view commitment, timing-aware issue, and retirement;
- **PuD compiler (later):** operation decomposition, dependencies, operand placement, and temporary rows;
- **scheduler (later):** selection among ready primitive commands;
- **functional model (later):** data-value correctness; and
- **energy model (later):** calibrated per-primitive energy.

This boundary preserves conventional DDR5 behavior and prevents placeholder electrical assumptions from becoming embedded in scheduling policy.

Table 6: Acceptance tests.

ID	Scenario	Required observation
S1	Open one row through each view.	Both rows remain open independently.
S2	Precharge one view.	Only that view closes.
S3	Open equal view-local line indices in A and B.	Both activations are accepted; equality of namespace-local labels creates no Milestone 1 intersection rule.
T1	Issue opposite-view activations.	Start times differ by at least `nXACTO`.
T2	Repeat T1 with an override.	The observed separation follows the configured value exactly when no stricter constraint applies.
T3	Issue `MRA_A`, then an A operation.	The A operation waits at least `nMRA`.
T4	Issue `MRA_A`, then a B operation.	B may start after cross-view constraints without waiting for the full `nMRA`.
T5	Exercise the four-activation window.	MRA contributes one provisional activation event.
P1	Issue MRA while its local view is open.	A local precharge is issued first; the other view remains open.
P2	Issue rank or same-bank maintenance.	Both views close before inherited maintenance executes.
C1	Run conventional DDR5 tests.	Command translation, timing, and state behavior are unchanged.
C2	Record a mixed A/B/MRA sequence.	Trace contains view identity and the expected timing dependencies.